Distinguishing Blockchain Forensics from Analytics - 30/1/25

BGIN Identity, Key Management & Privacy Working Group January 30, 2025

BGIN IKP Working Group Distinguishing Blockchain Forensics from Analytics	Forensics v Analysis 30th Jan.pptx
IKP-WG Meeting Agenda (2025-01-30)	Bitcoin Fog Case Summary.pptx
Website_Tentative_Program_BGIN_Block1 2.docx	Bitcoin Fog Case Daubert Hearing Summary
	IKP_Forensics or Analysis

Research Context

Background

- Growing need to distinguish between blockchain forensics and analytics
- Bitcoin Fog case highlighted critical questions about evidence standards
- Impact on legal proceedings and investigations
- Privacy and transparency implications

Key Questions

1. When does pattern analysis become admissible evidence?
2. How do we balance proprietary tools with transparency needs?
3. What constitutes proper forensic methodology in blockchain analysis?
4. ***How can we standardize practices while protecting innovation?***

Bitcoin Fog Case Study

Key Findings

- First major test of blockchain analytics in court (Daubert standard)
- Examination of clustering techniques and heuristics
- Validation of certain analytical methods
- Questions about transparency and reproducibility

Implications

- Sets precedent for evidence standards

- Highlights need for methodology documentation
- Raises questions about proprietary tools in legal proceedings
- Demonstrates the importance of peer review

Research Focus Areas

Technical Framework

- Evaluation of clustering techniques
- Analysis of heuristic methodologies
    - Difficult / private
- Data collection standards
- Verification procedures

Legal Standards

- Evidence admissibility criteria
- Expert witness requirements
- Documentation standards
- Cross-jurisdictional considerations

Privacy Considerations

- Balance between transparency and proprietary methods
- Data protection in forensic analysis
- Impact on individual privacy rights
- Ethical implications

Stakeholder Engagement

Key Participants

- Analytics Companies (Elliptic, Chainalysis, TRM Labs, etc.)
- Law Enforcement Agencies
- Legal Experts
- Academic Researchers
- Blockchain Companies
- Regulatory Bodies

Collaboration Methods

- Expert interviews
- Technical workshops
- Public consultations
- Peer review processes

Expected Outcomes

Deliverables

1. Framework Document
    - Clear definitions and distinctions
    - Methodological guidelines
    - Best practices recommendations
2. Standards Proposal
    - Evidence admissibility criteria
    - Tool validation requirements
    - Quality assurance procedures
3. Industry Guidelines
    - Practitioner certification requirements
    - Implementation recommendations
    - Risk management procedures

Next Steps

- Expert interviews and consultation
- Technical research continuation
- Case study analysis
- Draft framework development

Timeline

- January-March 2025: Research and consultation
- BGIN Block 12: Workshop and framework presentation
- Ongoing: Stakeholder feedback and refinement

Notes	Contributor
We should discuss the sharing of this paper and at what stages	MT

Discussion Points

Key Questions for Working Group

1. How can we better define the boundary between analytics and forensics?
2. What standards should apply to proprietary methodologies?
3. How can we ensure reproducibility while protecting IP?
4. What role should peer review play in validation?

Notes:

30/1/25

IKP WG

Cross jurisdictional - US centric case (different legal standards)

- Find other jurisdictions which have cases of using blockchain analysis
- Law enforcement officials
    - Different methods
    - Capacity / resource / access to tools
- Standardisation / guideline / rules discourse cross jurisdiction
    - Opening up the paper to global audience
- Canada Case -
    - Context [Dina]
- Other jurisdictions:
    - [chainalysis]
- BGIN role starting point in opening up the jurisdictional analysis
    - Crown prosecutors

https://www.ccn.com/dark-web-narcotrafficker-canada-18-million-bitcoin/

https://www.thestar.com/news/gta/province-seeks-forfeiture-of-1-4-million-bitcoin-wallet-taken-in-first-ever-seizure-by/article_474864c2-4bd0-5242-a041-6741f57e4c25.html

Difference between forensic techniques & the data or access to attribution data

- 'Attribution data is the pty tech'
    - Combination of offchain and onchain
    - Contexualising
-
- Interview questions:
    - Questions may be too broard (what is forensic vs analysis)
        - show some contextual information beforehand (interviewee educational material)

        - Build a fact file to make the discussion more concrete
        - [mitchell produce this]
    - Quality assurance
        - Technical (data science)
        - Practicioners
    - Conversation on the data science aspect
        - ‘Blockchain science’ fundamentals
        - Industry standards [eg: coinjoins]
    - Organisations describing what the data science
        - Services / products

Brief (one pager) paper on xyz, tech resources + high level

/ the core chain bases fundamentals

These conversations may give clarity on concepts / deriving inference from / facts / behavioural

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Blockchain Forensics vs Analytics Research

Expert Interview & Workshop Framework

Introduction (2 minutes)

Thank you for contributing to BGIN's research on blockchain forensics and analytics. This brief interview (20 minutes) will gather your initial insights, followed by your participation in our workshop at BGIN Block 12 where we'll explore these topics in depth with other experts.

Core Interview Questions (15-18 minutes)

Background & Experience (3-4 minutes)

1. Could you briefly describe your experience with blockchain analysis?
2. What's your perspective on the current state of blockchain forensics vs analytics?

Technical Insights (5-6 minutes)

Choose 2-3 most relevant to participant's background:

1. How do you distinguish between analytical tools and forensic evidence?
2. What validation processes do you consider essential?
3. How do you balance intellectual property protection with transparency needs?
4. What standards should apply to blockchain evidence in legal proceedings?

Future Development (5-6 minutes)

Choose 2-3 based on participant's expertise:

1. What key challenges need addressing in standardizing blockchain analysis?
2. How can we better validate forensic methodologies while protecting innovation?
3. What role should peer review play in blockchain forensics?
4. How should privacy considerations be balanced with forensic requirements?

Workshop Preparation (2 minutes)

1. Which aspects would you like to explore further in the Block 12 workshop?
2. Are there specific cases or examples you'd like to discuss?

BGIN Block 12 Workshop Participation

Workshop Format

- 90-minute interactive session
- Expert panel discussions
- Breakout groups on key topics
- Collaborative framework development

Workshop Topics

1. Forensics vs Analytics Distinction
    - Methodological differences
    - Validation requirements
    - Evidence standards
2. Technical Standards
    - Clustering techniques
    - Heuristic validation
    - Documentation requirements
3. Legal & Privacy Considerations
    - Court admissibility
    - Privacy protection
    - Cross-jurisdictional issues
4. Best Practices Development
    - Industry guidelines
    - Quality assurance
    - Training requirements

Post-Interview Notes

- Send workshop agenda and materials
- Share relevant research documents
- Confirm workshop participation details
- Note any specific topics for workshop focus

Confidentiality

- Attribution preferences for interview insights
- Workshop discussion ground rules
- Publication review opportunities

Expert Participation Invitation

BGIN Research Initiative: Distinguishing Blockchain Forensics from Analytics

Dear [Name],

Research Overview

The Blockchain Governance Initiative Network (BGIN) is conducting research on establishing clear distinctions between blockchain forensics and analytics. This work has become particularly relevant following the Bitcoin Fog case and growing industry discussions about evidence standards and methodological frameworks.

Why Your Participation Matters

Your expertise in [specific area] would provide valuable insights for this research. Recent developments in blockchain analysis have highlighted the need for clear standards and frameworks, particularly regarding [relevant aspect to their expertise].

Participation Options

We invite you to contribute in two ways:

1. Expert Interview (20 minutes)

- Brief discussion of key challenges and opportunities
- Focus on [specific area of expertise]
- Flexible scheduling between [dates]
- Remote participation via video call

2. BGIN Block 12 Workshop

- Date: [Workshop Date]
- Duration: 90 minutes
- Format: Interactive expert discussion
- Location: [Virtual/Physical location]
- Key topics:
    - Forensics vs analytics methodological distinctions
    - Evidence standards and validation
    - Privacy and transparency considerations
    - Best practices development

Expected Outcomes

Your participation will contribute to:

- Development of industry guidelines
- Framework for distinguishing forensics from analytics
- Standards for evidence admissibility
- Best practices recommendations

Next Steps

1. Please indicate your interest by replying to this email
2. Choose your preferred interview time slot: [Link to scheduling tool]
3. Confirm workshop participation preference

About BGIN

BGIN is a multi-stakeholder discussion body focused on blockchain governance. Our neutral stance and diverse expertise make us uniquely positioned to lead this research initiative.

Additional Information

- Research proposal: [Link]
- Current participants include experts from [general categories]
- Findings will be publicly shared through BGIN's channels
- Participation can be attributed or anonymous based on your preference

Contact

For any questions or to discuss further: [Contact details]

Looking forward to your participation in this important initiative.

Best regards,

[Your name] [Your title] BGIN Identity, Key Management & Privacy Working Group

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https://bgin-global.org/

https://bgin.discourse.group/

Blockchain Forensics vs Analytics Research

Insights Documentation Framework

Individual Interview Documentation

Participant Profile

```
Date: YYYY-MM-DD
Time: HH:MM
Duration: XX minutes
Name:
Organization:
Role:
Expertise Areas:
  - Area 1
  - Area 2
Attribution Preference: [Public/Anonymous/Partial]
Follow-up Permission: [Yes/No]
```

Key Insights Template

```
Technical Perspectives:
  Forensics Definition:
    Key Points:
      -
    Notable Quotes:
      -
    Examples Cited:
      -

  Analytics Definition:
    Key Points:
      -
```

```
    Notable Quotes:
      -
    Examples Cited:
      -

Methodology Insights:
  Validation Methods:
    Current Practices:
      -
    Challenges:
      -
    Suggestions:
      -

  Standards & Evidence:
    Current Approaches:
      -
    Pain Points:
      -
    Recommendations:
      -

Privacy Considerations:
  Concerns:
    -
  Solutions:
    -
  Trade-offs:
    -

Workshop Topics:
  Interest Areas:
    -
  Case Studies:
    -
  Discussion Points:
    -
```

Workshop Documentation

Session Information

```
Date: YYYY-MM-DD
Session: [Morning/Afternoon]
Duration: XX minutes
Total Participants: XX
Breakout Groups: XX
```

Discussion Themes

```
Theme 1: [Theme Name]
  Key Points:
    -
  Debates:
    -
  Consensus Areas:
    -
  Open Questions:
    -

Theme 2: [Theme Name]
  [Same structure as above]
```

Breakout Group Insights

```
Group: [Topic/Focus Area]
  Participants:
    - Role/Expertise
  Key Findings:
    -
  Challenges Identified:
    -
  Proposed Solutions:
```

```
    -
  Action Items:
    -
```

Cross-Cutting Analysis

Emerging Patterns

```
Common Themes:
  Theme:
    Frequency: XX mentions
    Context:
      -
    Supporting Quotes:
      -

Divergent Views:
  Topic:
    Perspective 1:
      -
    Perspective 2:
      -
    Key Tensions:
      -
```

Implementation Insights

```
Best Practices:
  Category:
    Current:
      -
    Proposed:
      -
    Challenges:
      -
```

```
Standards Development:
  Priority Areas:
    -
  Requirements:
    -
  Timeline Considerations:
    -
```

Follow-up Actions

```
Immediate:
  - Task
  - Owner
  - Timeline

Research Gaps:
  - Area
  - Potential Sources
  - Priority Level

Stakeholder Engagement:
  - Group
  - Next Steps

  - Timeline
```

Connect this framework to meeting and workshop transcripts*